Large Scale Magnetic Fields and The Turbulent Dynamo

Project

Submitted in partial fulfilment of the requirements of

IUCAA-NCRA Graduate School

By

Vaishak Prasad

Under the supervision of

Prof. Kandaswamy Subramanian

Inter-University Centre for Astronomy and Astrophysics

21 July 2017

Certificate

This is to certify that the Project entitled, "**Large Scale Magnetic Fields and the Turbulent Dynamo**" and submitted by MR VAISHAK P,a Project in partial fulfillment of the requirements of the IUCAA-NCRA Graduate School embodies the work done by him under my supervision and guidance.

Date: July 2017

(Supervisor):
Prof. Kandaswamy Subramanian
IUCAA

Contents

Contents **5**

1 Introduction **5**

2 Magnetic Fields in MHD **7**
2.1 The Induction Equation 7
2.1.1 Some Consequences 9
2.1.2 Flux Freezing 9
2.1.3 Magnetic Diffusion 11
2.1.4 Dissipation of Energy 11
2.2 Magnetic Helicity 12
2.2.1 Magnetic helicity and linkages of flux tubes 12
2.2.2 Conservation of Helicity 13

3 Dynamo Theory **15**
3.1 The Mean Field Dynamo 15
3.1.1 The Mean Field Approach 15
3.1.2 The Turbulent Dynamo equation 16
3.1.3 Turbulent transport coefficients 17
3.1.4 Additional contributions 18
3.1.5 Simple analytic Solutions to the α^2 Dynamo 19

4 Magnetic Helicity in the Mean Field Model **23**
4.1 The Large-scale magnetic helicity 23
4.2 The Small-scale Helicity 24
4.3 Conservation of net helicity 24
4.4 A Gauge invariant definition for helicity density 25
4.4.1 Helicity density through the Gauss Linking Formula 25
4.4.2 Double Fourier Transforms 26
4.4.3 Time evolution of the small-scale helicity density 27

5 Numerical Simulation of Dynamos **31**
5.1 Dynamo Action in Disk Galaxies 31
5.1.1 The Setup and the Equations 31

6 Summary **33**

A Appendix **35**
A.1 The MHD equations 35
A.1.1 The One Fluid Model 35

A.2 The Two fluid model 36
A.2.1 Deriving the MHD equation 36

Bibliography **39**

Acknowledgments

As always, all learning is only possible because of the Teachers: Through discussions and from books. I would like to acknowledge the continual support and guidance provided all along by my supervisor, Prof. Kandaswamy Subramanian. I am very grateful to him, who, as my Guide, has been the inspirational force all along, providing guidance and valuable advises through long discussions and clarification sessions. He has patiently spent time with me and persevered amidst my silly questions. He has corrected me when I went in the wrong direction and brought be back on the right path. I must say that I have greatly benefit learning under him.

I would like to thank the the Academic Faculty and the Non-Academic members of IUCAA for the resources and facilities extended. I am grateful to to many of my fellow researchers who have helped me at various points of the project. I would also like to thank them for questions and discussions helped me gain a better understanding of the subject.

ABSTRACT

In this project, we review the current understanding of the Turbulent Dynamo, that attempts to explain the observed large-scale coherence and strengths of the magnetic fields in galaxies. We begin with a brief overview of the current theory of Magnetohydrodynamics.We derive the Dynamo equation in the mean field approach and show how one calculates the turbulent emf in the First Order Smoothing Approximation.

We then continue on to the concept of Magnetic helicity and show how that it is a conserved quantity in ideal MHD. Then, it is introduced in the mean field approach. A gauge-invariant definition for the small-scale helicity as first given in a paper by K. Subramanian[1] and A. Brandenburg (2005)is presented. It is shown how one obtains various flux terms (in addition to the usual volume generation terms) in the evolution equation for the small-scale helicity density and how they can be calculated in terms of the double correlation functions. Since the action of large scale dynamos create linkages of fluxes in the small-scale field as a by-product, the dynamo is catastrophically quenched unless the small scale field can loose some helicity by means of its flux. The arguments involving these fluxes usually put forth to justify the efficient working of dynamos are also presented.

[1] Please see [1]

Chapter 1

Introduction

Observations suggest that all astrophysical objects are permeated by Magnetic fields. In particular, observations of Galaxies and Galaxy Clusters show that magnetic fields of the order of several micro-Gauss that are correlated on the scales of the sizes of these objects exist and are maintained by them (see, for instance fig 1.1). Furthermore, these magnetic fields in galaxies are also expected have influence on the physical processes going on inside them. Explaining the origin and existence of these fields is therefore one of the central problems of Plasma Astrophysics today.

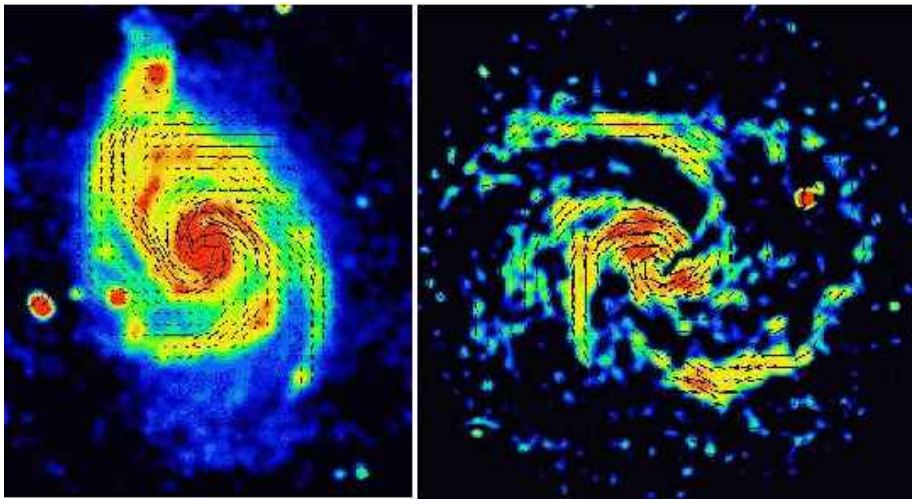


Figure 1.1: credit: K.Subramanian, A.Brandenburg [3]

Various theories exist that attempt to explain the properties of astrophysical magnetic fields :primarily their strength and large-scale correlation. The Turbulent Dynamo theory using the a Mean Field approach is one such candidate that explains how the turbulence in the fluid can exponentially amplify an initial seed field and how the magnetic fields can come to be correlated on such large-scales.

In this project, we review the current understanding of the Turbulent Dynamo theory through the Mean Field Approach beginning with the equations of Magnetohydrodynamics. We show how the arising turbulent transport coefficients (i.e. the mean turbulent emf) can be calculated in the simplest of the Approximations: The First Order Smoothing Approximation, resulting in the so-called so called α effect. Other contributions to the tubulent emf will be briefly mentioned. Then we discuss some well known simple analytic solutions to the Dynamo equation that exemplifies the possibility of exponential amplification of the mean fields.

The concept of Magnetic helicity, an important quadratic-invariant of ideal MHD

is discussed in the Mean field approach, in the context of the mean and the fluctuating field. A gauge-invariant definition of small-scale helicity density for weakly inhomogeneous turbulence[1] , and its time evolution is discussed in detail. Particularly, the expressions for various kinds of fluxes in this (weakly inhomogeneous) case and how one can calculate them have been portrayed. The importance of these fluxes in alleviating the catastrophic quenching of the α effect will also be discussed.

In the final section, a numerical code is setup to simulate the $\alpha\Omega$ dynamo in the context of Disk galaxies.

[1] see [1]

Chapter 2

Magnetic Fields in MHD

2.1 The Induction Equation

In plasmas, the magnetic field changes in time due to the production of electric fields which have a non-zero curl. These are produced by the currents that flow in them. The motion of the ions in fluids is self-supported, i.e. electromagnetic analogs to the case self-gravitating objects. The Flow of ions sets up currents, which source the electric and magnetic fields. These electric and magnetic fields act back on the fluids to alter the flow of ions through the Lorentz force.

In most cases, one is interested in the dynamics of the magnetic field in the plasma. One can derive an equation for the evolution of the Magnetic field by employing the Maxwell's equations (that describe how the flow affects the fields as) and the Ohms's Law (that tells how the fields affect the currents through Lorentz forces, taking into account the collisions in the fluid) s follows:

Consider the second of the Maxwell's equation:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.1}$$

and also consider the Ohm's Law :

$$\mathbf{J} = \rho_q \mathbf{V} + \sigma(\mathbf{E} + \mathbf{V} \times \mathbf{B}) \tag{2.2}$$

Here the first term is due to advection of the net charge ρ_q of the plasma.

The Electric field can be obtained in terms of the current density vector $\mathbf{J}$ from the Ohm's law, This has the form:

$$\mathbf{E} = \frac{\mathbf{J}}{\sigma} - \mathbf{V} \times \mathbf{B} - \frac{\rho_q \mathbf{V}}{\sigma}$$

This can exclusively be expressed in terms of the fields using the first and the last Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho_q}{\varepsilon_0}$$
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$

i.e. by substituting for ρ_q and $\mathbf{J}$ as

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$
$$\mathbf{J} = \frac{1}{\mu_0}\left(\nabla \times \mathbf{B} - \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}\right)$$

to get:

$$\mathbf{E} = \mathbf{V} \times \mathbf{B} + \eta(\nabla \cdot \mathbf{E})\mathbf{V} - \eta \nabla \times \mathbf{B} + \frac{\eta}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$

This equation contains some terms that directly contain the Fields and some that are in terms of derivatives of the fields. As one may naively guess, not all of these terms are equally important in all astrophysical situations. In fact, some of these terms are so small when compared to other terms that in almost every astrophysical context that they can be neglected. To understand this simply and to quantify this precisely, rewrite this equation by multiplying and dividing by c throughout and define $\eta = 1/\mu_0\sigma$ (so that dimensions and characteristic scales can be read off easily): , the equation has been written out by multiplying and dividing by factors of c so that dimensions and characteristic scales can be read off easily.

$$\mathbf{E} = \frac{\mathbf{V}}{c} \times (\mathbf{cB}) + \frac{\eta}{c}(\nabla \cdot \mathbf{E})\frac{\mathbf{V}}{c} - \frac{\eta}{c}\nabla \times (\mathbf{cB}) + \frac{\eta}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$

The last term will be small when compared to the first term i.e. the electric field when the electric field varies on a time-scale T much larger than the Faraday time-scale defined through $\tau_F = \eta/c^2$.In these circumstances, the displacement current term can be neglected. Further, the term involving the spatial derivative of **E** can also be neglected when the length-scale over which **E** varies is much larger than the corresponding Faraday length-scale times the velocity factor.Here V is the characteristic bulk velocity of the fluid.

$$T >> \tau_F = \frac{\eta}{c^2} \qquad\qquad L >> (\tau_F c)\frac{v}{c} = \tau_F V. \tag{2.3}$$

By substituting back the Electric field from the above equation into the first equation (2.1), one arrives at an equation exclusively for the magnetic field:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \left(\mathbf{V} \times \mathbf{B} + \varepsilon_0 \frac{(\nabla \cdot \mathbf{E})\mathbf{V}}{\sigma} - \frac{\nabla \times \mathbf{B}}{\mu_0 \sigma} + \frac{1}{c^2 \mu_0 \sigma}\frac{\partial \mathbf{E}}{\partial t}\right)$$

However, when the above conditions (2.3) are satisfied the approximations lead to a simple equation for the evolution of the magnetic field:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B} - \eta \nabla \times \mathbf{B}) \tag{2.4}$$

The quantity $\eta = 1/(\mu_0\sigma)$ defined above is called as the 'Diffusivity'. This equation is known as the *Induction Equation*

Some points to note here are,

- The term due to $\mathbf{V} \times \mathbf{B}$ is called the *Induction term* and that due to $\nabla \times \mathbf{B}$ is called the *Diffusion term* (further elaborated below). The relative importance of the induction term to the diffusion term is quantified by the *Magnetic Reynolds Number* R_m:

$$R_m \equiv \frac{V L_{sys}}{\eta}$$

Where V is a characteristic velocity of the fluid in the plasma and L_{sys} is a characteristic length-scale in the system. In most cases of Astrophysical interest, R_m is large with $R_m > 30$

In the following section, a general description/ a discussion of the features of this equation will be given.

2.1.1 Some Consequences

Some well known ideas concerned with the Induction equation are those of (Approximate) Flux Freezing and the Diffusion of the Magnetic field.

2.1.2 Flux Freezing

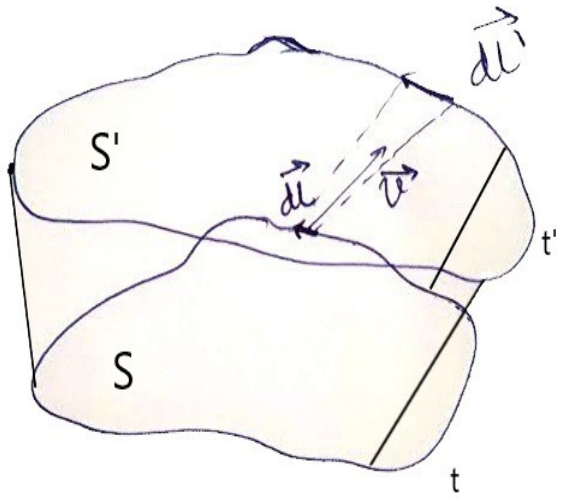


Figure 2.1: The evolution of the boundary of the area S with time as the fluid flows.

Consider the Magnetic flux passing through an area element S in the fluid (see fig 2.1). As time evolves, the fluid elements associated with the area A move, changing the area A. The magnetic field also changes alongwith, bringing about a change in the flux passing through that part of the fluid. The net change in the magnetic flux can thus be written as a sum of two pieces:

$$\frac{d}{dt}\int_S d\mathbf{S}\cdot\mathbf{B} = \int_A d\mathbf{S}\cdot\frac{\partial\mathbf{B}}{\partial t} + \oint_{\partial A}(\mathbf{V}\times\mathbf{dl})\cdot\mathbf{B}$$
$$= \int_S d\mathbf{S}\cdot\left(\frac{\partial\mathbf{B}}{\partial t} + \nabla\times(\mathbf{B}\times\mathbf{V})\right)$$

Now consider the limit $\eta \to 0$ in the induction equation (2.4). In this case, the last term of the drops out (provided the currents remain finite) and the equation now says that the above expression on the LHS i.e. the time derivative of the flux through the surface S is zero:

$$\frac{d}{dt}\int_S d\mathbf{S}\cdot\mathbf{B} = \int d\mathbf{S}\cdot\left(\frac{\partial\mathbf{B}}{\partial t} - \nabla\times(\mathbf{V}\times\mathbf{B})\right) = 0$$

Another way to express the same thing is by expanding out the $\nabla\times(\mathbf{V}\times B)$ term in the magnetic induction equation using the vector identity for $\nabla\times(\mathbf{A}\times B)$:

$$\nabla\times(\mathbf{A}\times B) = \mathbf{B}\cdot\nabla\mathbf{A} - \mathbf{A}\cdot\nabla\mathbf{B} + \mathbf{A}\nabla\cdot\mathbf{B} - \mathbf{B}\nabla\cdot A \tag{2.5}$$

and recalling that $\nabla\cdot\mathbf{B} = 0$ to write:

$$\frac{\partial\mathbf{B}}{\partial t} = (\mathbf{B}\cdot\nabla)\mathbf{V} - (\mathbf{V}\cdot\nabla)\mathbf{B} - (\nabla\cdot\mathbf{V})$$

Or,

$$\frac{d}{dt}\left(\frac{\mathbf{B}}{\rho}\right) = \left(\frac{\mathbf{B}}{\rho}\right)\cdot\nabla\mathbf{V} \tag{2.6}$$

where d/dt is the total (Lagrangian) derivative : $d/dt = \partial/\partial t + \mathbf{V}\cdot\nabla$

Let us now consider the time evolution of infinitesimal separation vector $\delta\mathbf{x}_{ab} = \mathbf{x_b} - \mathbf{x_a}$ separating two neighboring fluid elements 'a' and 'b' in the Lagrangian description:

$$\frac{d}{dt}\delta\mathbf{x}_{ab}(t) = \mathbf{V_b}(t) - \mathbf{V_a}(t) = \delta\mathbf{x}_{ab}(t)\cdot\nabla\mathbf{V}$$

Or, in general,

$$\frac{d}{dt}\delta\mathbf{x}(t) = \delta\mathbf{x}_(t)\cdot\nabla\mathbf{V}$$

This exercise shows that $\mathbf{B}/\rho$ follows the same evolution equation as that of the infinitesimal separation vector δx between two neighboring fluid elements. Hence, the solution of the above equation (2.6) can be expressed as:

$$\left(\frac{\mathbf{B}}{\rho}\right) = C\delta\mathbf{x}$$

This clearly portrays the behavior of $\mathbf{B}/\rho$ as the fluid is stretched / compressed; e.g., in an incompressible case, in a flux tube of length l and cross section S containing an initial field of strength of B (say, by differential rotation leading to a λ fold increase in the length l) , the magnetic field after elongation will get amplified to a strength λB. Also, the amplification is linear in the amount of stretch i.e. the fields are amplified on a linear time scale in the presence of shear.

This constraint viz. (approximate) Flux Freezing of ideal MHD is then one of the means by which the Magnetic fields can be amplified from a given initial seed field.

One can estimate the timescale that would be required by the above effect i.e. assuming linear growth, to grow the magnetic fields in galaxies from an initial seed strength (from accepted models) of around 10^{-19}G to the present day strength of around μG. This gives a timescale $\tau \sim 10^{28}$s , a timescale greater than the age of the universe. This is one of the most important questions of Astrophysics, i.e. how Magnetic fields in galaxies and clusters have come to assume such high field strengths and how they came to be correlated on large-scales,scales of the order of the size of the astrophysical system.

2.1.3 Magnetic Diffusion

For very small R_m, the last term of the induction equation (2.4) becomes dominant. On expanding the curl, the equation can be written as:

$$\frac{\partial \mathbf{B}}{\partial t} = \eta \nabla^2 \mathbf{B}$$

Which is a diffusion equation for the Magnetic field **B**.

The content of the induction equation (2.4) can now be naively put as the constraint of flux freezing along with the diffusion of Magnetic field.

2.1.4 Dissipation of Energy

Consider the time rate of change of energy stored in Magnetic fields:

$$\frac{\partial}{\partial t}\int_V \frac{1}{2\mu_0}\mathbf{B}^2\, dV = \frac{1}{\mu_0}\int_V \mathbf{B}\cdot\frac{\partial \mathbf{B}}{\partial t}\, dV$$

Using the same vector identity (2.5) to expand $\nabla \times (\mathbf{E}\times\mathbf{B})$ and substituting for the electric field from Ohm's law, we can re-write the above equation as:

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \frac{1}{2\mu_0}\mathbf{B}^2\, dV &= -\frac{1}{\mu_0}\int_V (\nabla\cdot(\mathbf{E}\times\mathbf{B}) + \mathbf{E}\cdot\mathbf{J})\, dV \\ &= -\eta\int_V \mathbf{J}^2\, dV - \frac{1}{\mu_0}\int_V \mathbf{V}\cdot(\mathbf{J}\times\mathbf{B})\, dV - \frac{1}{\mu_0}\int_{\partial V}(\mathbf{E}\times\mathbf{B})\cdot d\mathbf{S}\end{aligned} \tag{2.7}$$

The first term proportional to the squared of the current density is the usual dissipation due to resistivity of the plasma. The second one is the power delivered by the Lorentz force to the plasma and the last one is loss of energy due to Poynting flux.

One can also calculate the time rate of change of kinetic energy of the plasma:

$$\frac{\partial}{\partial t}\int_V \rho_m \mathbf{V}^2\, dV = \int_V p\nabla\cdot\mathbf{V}\, dV + \frac{1}{\mu_0}\int_V \mathbf{V}\cdot(\mathbf{J}\times\mathbf{B})\, dV + \int_V \mathbf{V}\cdot\mathbf{g}\, dV - 2\nu\int_V \rho_m [S]^2\, d^3\mathbf{r}$$

Here the first term is the work done by pressure due to compression/expansion of the fluid, the second term is the work done by Lorentz force, the third due to the gravitational field $g = -\nabla\phi$ and the last involving the traceless rate of strain tensor is taking into account the dissipation due to viscosity.

2.2 Magnetic Helicity

There are some quadratic quantities that are conserved by the flow i.e. nonlinear evolution of the fluid in ideal MHD (when the plasma is infinitely conductive). These serve as important constraints and prove useful in various theories of dynamos. One of these is called *Magnetic Helicity* ,defined as:

$$H = \int_V \mathbf{A}\cdot(\nabla\times\mathbf{A})\, dV = \int_V \mathbf{A}\cdot\mathbf{B}\, dV \tag{2.8}$$

Where the integration is over the volume V of the system. The integrand in the above expression is the *Magnetic Helicity Density*, given by $\mathbf{A}\cdot\mathbf{B}$. The magnetic helicity gives a measure of the number of linkages of magnetic flux tubes, as will be explained later in this section.

Since the Magnetic helicity involves the vector potential explicitly, one may guess that the quantity is not gauge invariant in general. Nonetheless it is an important quantity in Dynamo theory as explained below. As for the Gauge dependence, it can be shown that the magnetic helicity (2.8) is gauge invariant under specific conditions:

Consider the Gauge transformation of the Vector Potential $\mathbf{A}\to\mathbf{A}+\nabla\chi$. The Total helicity (2.8) modifies to:

$$H' = H + \int_V (\nabla\chi\cdot\mathbf{B})dV$$

The last term can be re written as $\nabla\cdot(\chi\mathbf{B})$. Then the additional contribution to H can be expressed as an integral over the boundary surface of the system:

$$H' - H = \int_{\partial V} (\chi\mathbf{B})\cdot\, d\mathbf{S}$$

The Magnetic helicity (2.8) will thus be gauge invariant if the magnetic field is tangent to the boundary of the system or vanishes on the boundary.

2.2.1 Magnetic helicity and linkages of flux tubes

One can demonstrate that two loops of magnetic flux tubes have finite helicity if they are linked (see fig 2.2. For the configuration depicted in the figure, assume that the flux tubes are infinitesimally thin. Under this simplifying assumption, we can work out the Magnetic Helicity as defined by (2.8):

$$H = \int_V (\mathbf{A}\cdot\mathbf{B})\, dV = \Phi_A\oint_{\partial A}\mathbf{A}\cdot\mathbf{dl} + \Phi_B\oint_{\partial B}\mathbf{A}\cdot\mathbf{dl}$$

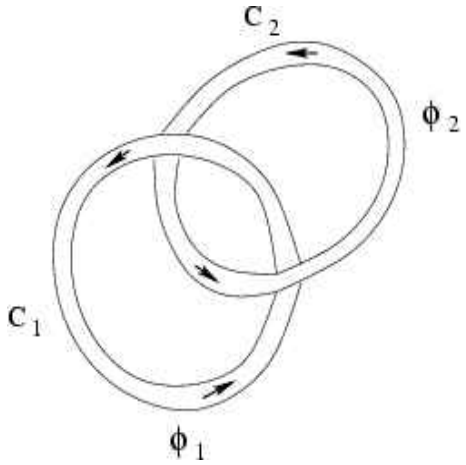


Figure 2.2: Two linked infinitesimally thin flux tubes. The magnetic field is anticlockwise in both the loops in this orientation Credit: K.Subramanian, Brandenburg 2005[3]

Where each piece corresponds to a closed integral around each of the (Infinitesimally thin) flux tubes. Here, Φ_A and Φ_B are the magnetic fluxes contained in each of the flux tubes.

Also, by converting the above line integrals into area integrals through the Stokes theorem:

$$H = \Phi_A \oint_A \nabla \times \mathbf{A} \cdot \mathbf{dS} + \Phi_B \oint_B \nabla \times \mathbf{A} \cdot \mathbf{dS}$$

Which reduces to $2\Phi_A\Phi_B$. Thus, we can write

$$H = +2\Phi_A\Phi_B$$

Note that in the figure 2.2, the magnetic field is *anti-clockwise* in both the loops and hence the $+$ sign. If it were to be clockwise in both instead, the net Helicity would have been taken the minus sign. And in the case where it was clockwise in one and anticlockwise in the other, the net helicity would have turned out to be zero. The Magnetic helicity, as can be seen from its defining equation (2.8), is a *pseudoscalar*

This simple illustration exemplifies the situation where one would have finite non-zero Magnetic helicity. Owing to its simplicity, one can interpret the net helicity of a system as a measure of the strength of linkages of the flux tubes.

2.2.2 Conservation of Helicity

As mentioned earlier, quantities like the Magnetic helicity are important because these are approximately conserved in resistive MHD. Consider the time evolution of the total Magnetic Helicity:

$$\frac{dH}{dt} = \int_V \frac{\partial \mathbf{A}}{\partial t} \cdot \mathbf{B}\, dV + \int_V \mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial t}\, dV$$

Using $\frac{\partial \mathbf{A}}{\partial t} = -\nabla\phi - \mathbf{E}$ and Faraday's Law we have

$$\begin{aligned}\frac{\partial H}{\partial t} &= \int_V (-\nabla\phi \cdot \mathbf{B} - \mathbf{E} \cdot \mathbf{B} - \mathbf{A} \cdot \nabla \times \mathbf{B})\, dV \\ &= \int_V (-\nabla \cdot (\phi\mathbf{B} + \mathbf{A} \times \mathbf{E}) - 2\mathbf{E} \cdot \mathbf{B}) dV\end{aligned}$$

Using the Ohm's Law, one can substitute for the electric field as $\mathbf{E} = \mathbf{J}/\sigma - \mathbf{V} \times \mathbf{B}$ and simplify the above expression to:

$$\frac{\partial H}{\partial t} = -2\eta \int_V (\mathbf{B} \cdot \nabla \times \mathbf{B})\, dV = -2\eta \int_V (\mathbf{J} \cdot \mathbf{B})\, dV \tag{2.9}$$

One can compare this with the equation governing the dissipation of energy stored in the Magnetic Field (2.7):

Since the rate of change of net Helicity (2.9) is only proportional to $\mathbf{J} \cdot \mathbf{B}$ while the energy decays as $\mathbf{J}^2$, the helicity of the system decays on a slower timescale than the energy stored in the Magnetic fields even in the case of non zero resistance.

As discussed earlier in 2.1.2, a linear growth does not suffice to explain the strength of the observed magnetic fields. Moreover, there must be a mechanism that operates to sustain the magnetic fields in the in the presence of dissipation. The above equation shows that the energy in the magnetic fields can be increased if work can be done against the Lorentz force. In such a process, the kinetic energy of the plasma is converted into electromagnetic energy. Such a mechanism is called a *Dynamo*, the subject of the next chapter.

Chapter 3

Dynamo Theory

As discussed in the previous chapter, the linear growth of magnetic fields is insufficient to account for their strengths observed in astrophysical objects today. Besides, the large scale coherence of the fields are also to be explained.

The large scale fields in galaxies are maintained by their differential rotation and turbulence. Due to turbulence, the motion of matter in galaxies have a random fluctuating component apart from the bulk motion of the fluid. The amplification of the magnetic field and its sustenance is achieved by the conversion of the turbulent kinetic energy into electromagnetic energy.

The Turbulent Dynamo is a mean field approach that provides a way to achieve this. In this chapter we will see that helical turbulence can support and maintain large-scale magnetic fields.

3.1 The Mean Field Dynamo

3.1.1 The Mean Field Approach

In the Mean field approach, all quantities are expressed as a sum of a Mean Field quantity, something that varies on a length scale much larger than the correlation scale of turbulence, and a fluctuating part (a random field) that varies on a smaller scale. In the Mean field approach, one writes:

$$\begin{aligned} \mathbf{B} &= \overline{\mathbf{B}} + \mathbf{b} \\ \mathbf{V} &= \overline{\mathbf{V}} + \mathbf{v} \end{aligned} \tag{3.1}$$

Here overbars denote averaging over some relevant length-scale that is larger than the turbulent scale. throughout, the corresponding small letters will be used to denote the random small-scale component whose mean is assumed to be zero. i.e. for instance $\overline{\mathbf{v}} = 0 = \overline{\mathbf{b}}$.A similar decomposition will be assumed for all the fluid quantities when necessary.

One then attempts to derive equations governing the mean and fluctuating fields. For instance, in this case, a corresponding set of Maxwell's equations can then be written down separately for the dynamics of the random-small scale component by subtracting the Maxwell's equation written down for the mean electric and magnetic fields (which are just the usual set of Maxwell's equations (A.1) with overbars denoting averages) from the Maxwell's equations for the total field, which are:

$$\nabla \cdot \mathbf{e} = \frac{1}{\varepsilon_0}\delta\rho_q$$
$$\nabla \times \mathbf{e} = -\frac{\partial \mathbf{b}}{\partial t}$$
$$\nabla \cdot \mathbf{b} = 0$$
$$\nabla \times \mathbf{b} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{e}}{\partial t}$$

For the case of Astrophysical plasmas, the fluctuating part is largely due to *turbulence*, while the mean field component attempts to describe the Magnetic field that is correlated on a large-scale.

3.1.2 The Turbulent Dynamo equation

On substituting the above decomposition (3.1) into the Induction equation (2.4), taking average of the equation and assuming the mean and the fluctuating components are independent, we have:

$$\frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B} + \overline{\varepsilon} - \eta \nabla \times \mathbf{B}) \tag{3.2}$$

for the mean field and

$$\frac{\partial \mathbf{b}}{\partial t} = \nabla \times (\overline{\mathbf{V}} \times \mathbf{b} + \mathbf{v} \times \overline{\mathbf{B}} - \overline{\mathbf{v} \times \mathbf{b}} + \mathbf{v} \times \mathbf{b} - \nabla \times \mathbf{b}) \tag{3.3}$$

for the fluctuating field. The quantity $\overline{\boldsymbol{\mathcal{E}}} = \overline{\mathbf{v} \times \mathbf{b}}$ is called the turbulent *mean electromotive force* or *turbulent emf* in short.

In order to understand the Turbulent Dynamo, one would ideally want to solve the the mean field dynamo equation and understand its solutions. However in order to solve the equation, as a characteristic of all Mean field theories, one has to first find a way to express the terms involving correlation of the fluctuating fields (in this case, it is the emf $\overline{\varepsilon}$) in terms of the Mean field quantities i.e. one has to employ a certain closure theory. Such a general scheme is almost always hard to come by, but nevertheless exist for cases involving simplifying assumptions.

For this case, one usually writes the mean electromotive force term $\overline{\varepsilon}$ as a series expansion containing different orders of spatial derivatives of the Magnetic field:

$$\overline{\boldsymbol{\mathcal{E}}}_i = \alpha_{ij}\overline{B}^j + \beta_{ijk}\frac{\partial \overline{B}^j}{\partial x_k} + \gamma_{ijkl}\frac{\partial \overline{B}^j}{\partial x_k \partial x_l} + ...$$

The coefficients multiplying the derivatives of the component of mean magnetic field $\alpha, \beta, \gamma, ..$ are known as the *Turbulent transport coefficients* and are *pseudo-tensors*. This is because, the magnetic field itself is a pseudo-vector and this implies that the mean emf $\overline{\boldsymbol{\mathcal{E}}}$ that involves a cross product with the magnetic field is a vector. Therefore, the coefficients multiplying the derivatives of the pseudo-vector $\mathbf{B}$ on the right hand side must be pseudo -tensors. This will be discussed in the upcoming section.

3.1.3 Turbulent transport coefficients

Fist Order Smoothing Approximation (FOSA)

One of the simplest schemes used for calculating the mean emf (or the turbulent transport coefficients) is known by the name *First Order Smoothing Approximation*.

The mean emf is calculated as follows:

In the equation for the fluctuating part of the field (3.3), one first neglects the term $\overline{\mathbf{v}\times\mathbf{b}}-\mathbf{v}\times\mathbf{b}$ postulating that the quadratic fluctuations are very small. This is known as the *First Order Smoothing Approximation*, which neglects fluctuations of quadratic and higher order. Then, taking a cross product of the equation with the second equation (3.3) and integrating over time, we have:

$$\overline{\boldsymbol{\mathcal{E}}} = \mathbf{v}(t)\times\left(\nabla\times\int_0^t \mathbf{v}(\mathbf{t}')\times\overline{\mathbf{B}}(t')\,dt'\right)$$

In index notation, one can simplify this to:

$$\overline{\boldsymbol{\mathcal{E}}}_i = \varepsilon_{ijk}\varepsilon_{klm}\int_0^t \varepsilon_{mpq}\,\left(\overline{v_j(t)\,\partial_l(v_p(t'))}\,\overline{B}_q(t') + \overline{v_j(t)v_p(t')}\,\partial_l\overline{B}_q(t')\right)$$

Using the identity $\varepsilon_{klm}\varepsilon mpq = \delta_{kq}\delta lp - \delta_{kp}\delta lq$, and the divergence free conditions for the magnetic and random small-scale velocity $\nabla\cdot\mathbf{b}=0=\nabla\cdot\mathbf{v}$ the above expression can be simplified to:

$$\begin{aligned}\overline{\boldsymbol{\mathcal{E}}}_i &= \varepsilon_{ijk}\left(\int_0^t \overline{v_j(t)v_p(t')}\partial_p\overline{B}_k(t')\,dt' - \int_0^t \overline{v_j(t)\partial_p(v_k(t'))}\overline{B}_p(t')\,dt'\right)\\ &= \left(\int_0^t \alpha_{ip}(t,t')\overline{B}_p(t')\,dt' - \int_0^t \eta_{ilp}(t,t')\partial_l\overline{B}_p(t')\,dt'\right)\end{aligned}$$

Where

$$\alpha_{ip} = \varepsilon_{ijk}\overline{v_j(t)\partial_p(v_k(t'))} \qquad\qquad \eta_{ilp} = \varepsilon_{ijp}\overline{v_j(t)v_l(t')}$$

are called as the transport coefficients. If one assumes that the turbulence is statistically stationary, then the correlation functions depend only upon the difference in time $(t-t')$. Furthermore, if one assumes that these coefficients are isotropic due to the nature of turbulence, then one can obtain a simple expression for these coefficients.

Due to isotropy, we can write

$$\alpha_{ip} = C_1\delta_{ip} \qquad\qquad \beta_{ilp} = C_2\varepsilon_{ilp}$$

Taking the trace of α and Multiplying β by ε_{ijk} and then taking the trace, one gets:

$$C_1 = -\frac{1}{3}\overline{\mathbf{v}(t)\cdot\omega(t')} \qquad\qquad C_2 = \frac{1}{3}\overline{\mathbf{v}(t)\cdot\mathbf{v}(t')}$$

Finally, if the correlation timescales are small, over which the mean magnetic field **B** does not vary, the integral over time can be effectively achieved by multiplying the

integrand by the correlation time τ_c. Defining

$$\alpha = -\frac{1}{3}\overline{\mathbf{v}(t)\cdot\boldsymbol{\omega}(t')}\tau_c$$
$$\eta = \frac{1}{3}\overline{\mathbf{v}(t)\cdot\mathbf{v}(t')}\tau_c$$

the expression for the mean electromotive force $\overline{\mathscr{E}}$ can be simply written as:

$$\overline{\boldsymbol{\mathcal{E}}} = \alpha\overline{\mathbf{B}} - \eta_t\nabla\times\overline{\mathbf{B}}$$

The first term proportional to $\mathbf{B}$ is the term responsible for what is known as the *Alpha Effect* and gives rise to the α Dynamo which will be discussed in the subsequent sections. The second term adds to the Diffusion term of the usual induction equation (2.4), enhancing the diffusivity. Its coefficient η_t is known as the *turbulent diffusivity*, the turbulent contribution to the usual microphysical diffusivity. Its value is usually much larger than its microphysical counterpart i.e. $\eta_t >> \eta$. Usually the total diffusivity is denoted by $\eta_T = \eta_t + \eta$.

Substituting this into the Mean Field Dynamo equation (3.2), we have the:

$$\frac{\partial\overline{\mathbf{B}}}{\partial t} = \nabla\times(\mathbf{V}\times\mathbf{B} + \alpha\mathbf{B} - \eta_T\nabla\times\mathbf{B})$$

Some of the names referring to dynamo action when specific terms become important in the dynamo equation are as follows:

- α^2 Dynamo: When thefirst term of the dynamo equation involving velocity $\overline{\mathbf{V}}$ is negligible in magnitude relative to the α term in the dynamo equation. This also means that after the generation of the toroidal fields from the poloidal fields through the α effect, the poloidal fields are in-turn regenerated by the toroidal fields themselves through same α effect. The α effect is thus mainly responsible for the the sustenance of the large-scale magnetic fields.
- $\alpha\omega$ Dynamo: The dynamo action when the α term is negligible in comparison to the shearing term in the dynamo equation for toroidal component. In effect, this means that the dynamo action through the shearing effect is majorly responsible for the regeneration of the poloidal fields rather than the α effect.
- $\alpha^2\omega$ Dynamo: This dynamo action when all effects are equally important.

3.1.4 Additional contributions

In calculating the above transport coefficients only the contribution from the $\mathbf{v}\times\overline{\mathbf{B}}$ in the equation for the small scale magnetic field (i.e. (3.3)) was considered. The contribution from the term $\overline{\mathbf{V}}\times\mathbf{b}$ can also be calculated in exactly the same manner in the FOSA, leading to similar coefficients involving double correlations. The dynamo equation when written down including these contributions takes the form:

$$\frac{\partial\overline{\mathbf{B}}}{\partial t} = \nabla\times(\mathbf{V}\times\mathbf{B} + \alpha\mathbf{B} - \mu\overline{\mathbf{V}} + \gamma\overline{\boldsymbol{\omega}} - \eta_T\nabla\times\mathbf{B})$$

Where the new coefficients are given by:

$$\mu = -\frac{1}{3}(\overline{\mathbf{v}\cdot\nabla\times\mathbf{b}})\ \tau_c$$
$$\gamma = \frac{1}{3}(\overline{\mathbf{v}\cdot\mathbf{b}})\ \tau_c$$

These new contributions turn out to not be extremely small as compared to the other two as the cross helicities are usually orders of magnitude smaller than the kinetc helicity. Also to note is that Galilean invariance may be lost in case the term proportional to **V** is retained.

3.1.5 Simple analytic Solutions to the α^2 Dynamo

Analytic solutions to the Dynamo equation are difficult to obtain in general. However, solutions or a qualitative description of their behavior have been derived for specific cases which involve some simplifying assumptions regarding the nature of the solutions. A mention of two of these, one qualitative, (by no means non-exhaustive) can be found in this section.

The α^2 Dynamo

The α^2 Dynamo equation is given by:

$$\frac{\partial\overline{\mathbf{B}}}{\partial t} = \nabla\times(\alpha\mathbf{B} - \eta_T\nabla\times\mathbf{B}) \tag{3.4}$$

For the case of constant α and the total diffusivity η_T, consider taking the curl of this expression on both the sides:

$$\nabla\times\frac{\partial\overline{\mathbf{B}}}{\partial t} = \nabla\times\nabla\times(\alpha\mathbf{B} - \eta_T\nabla\times\mathbf{B})$$

Or

$$\frac{\partial}{\partial t}\nabla\times\overline{\mathbf{B}} = \nabla\times(\alpha(\nabla\times\overline{\mathbf{B}}) - \eta_T\nabla\times(\nabla\times\overline{\mathbf{B}}))$$

This suggests that, if $\overline{\mathbf{B}}$ is a solution to the dynamo equation, then so is $\nabla\times\overline{\mathbf{B}}$. Now the simplest case when this can happen is when $\nabla\times\overline{\mathbf{B}} = k\overline{\mathbf{B}}$, where k is assumed to be a real constant. Proceeding with this assumption, one arrives at an eigen-value problem for the time evolution of $\overline{\mathbf{B}}$:

$$\frac{\partial\overline{\mathbf{B}}}{\partial t} = (\alpha k - \eta_T k^2)\overline{\mathbf{B}}$$

which has solutions that are exponential in time. Assuming $\overline{\mathbf{B}}(\mathbf{x},t) = \overline{\mathbf{B_0}}(\mathbf{x})e^{\gamma t}$, one arrives at the following quadratic equation:

$$\eta_T k^2 - \alpha k + \gamma = 0$$

the solutions of which are:

$$k = \frac{\alpha \pm \sqrt{\alpha^2 - 4\eta_T \gamma}}{2\eta_T}$$

This equation has to be supplemented with boundary conditions, which will give another relation between the constants γ and k, fixing them.

Thus for these simplifying assumptions, one has solutions that are either decaying/growing. Specifically, growth occurs for small enough scales when $\lambda > \lambda_{cr}$ where λ_{cr} is the scale for which $\gamma = 0$ i.e. $\lambda = \alpha/\eta$. The large-scales defined by $\lambda > \lambda_{cr}$ decay.

Solution by means of a Complex transformation

Another way to physically obtain the above can be solved is in the context where the α^2 dynamo equation for the x and y is written down in Cartesian coordinates:

$$\begin{aligned}\frac{\partial \overline{B}_x}{\partial t} &= -\alpha \frac{\partial \overline{B}_y}{\partial z} + \eta_T \frac{\partial^2 \overline{B}_x}{\partial z^2} \\ \frac{\partial \overline{B}_y}{\partial t} &= +\alpha \frac{\partial \overline{B}_x}{\partial z} + \eta_T \frac{\partial^2 \overline{B}_y}{\partial z^2}\end{aligned} \tag{3.5}$$

This is applicable to a disk geometry where the thickness of the disk is very small in comparison to its radial extent, a situation relevant in disk galaxies. This implies that the radial derivatives are vary small when compared to the derivatives along the height (i.e. z derivatives). Due to this, one make a local coordinate transformations to cylindrical polar co-ordinates and preserve the same form . These equations thus give the local evolution of the mean magnetic field.

Due to the symmetry of this equation, one can describe the time evolution of the radial and azhimuthal components in terms of one complex field defined by:

$$\tilde{B} = \overline{B}_x + i\overline{B}_y$$

The equation can be obtained by multiplying the second of the above set (3.5) by the imaginary complex number i and adding the two:

$$\frac{\partial \tilde{B}}{\partial t} = i\alpha \frac{\partial \tilde{B}}{\partial z} + \eta_T \frac{\partial^2 \tilde{B}}{\partial z^2} \tag{3.6}$$

The above equation can be easily solved using Fourier analysis.

Furthermore, consider the vector potential in this description. Writing $\overline{\mathbf{B}} = \nabla \times \overline{\mathbf{A}}$ in the α^2 dynamo equation (3.4):

$$\frac{\partial}{\partial t}(\nabla \times \overline{\mathbf{A}}) = \nabla \times (\alpha(\nabla \times \overline{\mathbf{A}}) - \eta_T \nabla \times (\nabla \times \overline{\mathbf{A}}))$$

Which can be written as:

$$\nabla \times \left(\frac{\partial \overline{\mathbf{A}}}{\partial t} - \nabla \times (\alpha \overline{\mathbf{A}} - \eta_T \nabla \times \overline{\mathbf{A}}) \right) = 0$$

This implies that the expression inside the bracket can be expressed as a curl free function $-\nabla\chi$:

$$\frac{\partial \overline{\mathbf{A}}}{\partial t} - \nabla \times (\alpha \overline{\mathbf{A}} - \eta_T \nabla \times \overline{\mathbf{A}}) = -\nabla\chi$$

This is a manifestation of the gauge freedom in the choice of the magnetic vector potential $\overline{\mathbf{A}}$. Thus in a suitable gauge, the mean magnetic vector potential also follows the same equation as that for the magnetic field:

$$\frac{\partial \overline{\mathbf{A}}}{\partial t} - \nabla \times (\alpha \mathbf{A} - \eta_T \nabla \times \overline{\mathbf{A}}) = 0$$

Now consider plane wave solutions to the complex equation for $\tilde{B}$ i.e. solutions of the form $\tilde{B} = B_0 e^{ikz-\omega t}$, where ω and k are allowed to be complex. Substituting this into the equation (3.6) The condition on k that is obtained is:

$$\eta_T k^2 + \alpha k - i\omega = 0$$

which has the solutions:

$$k = \frac{-\alpha \pm \sqrt{\alpha^2 + 4i\eta_T \omega}}{2\eta_T}$$

This method was also used in [4] to show that one can have oscillatory solutions to the α^2 dynamo equation. See for further details.

Chapter 4

Magnetic Helicity in the Mean Field Model

In section 2.2, we saw that one can define a quantity called Magnetic Helicity that is a measure of the degree of linkages of magnetic flux tubes (2.8). Similarly, in the mean field approach also, one can define an analogous quantity for both the mean and the random small scale field. In this section, volume integrated helicities for the large-scale and small scale fields would be discussed, followed by their time evolution. This would then naturally lead us to the requirement of a gauge-invariant quantity for the small-scale magnetic helicity density, a quantity that would give a measure of the density of linkages of flux tubes in a gauge invariant fashion. Then, a discussion of its time evolution would follow, leading to terms that denote *fluxes* of the small-scale helicity. It will be argued that such fluxes can be useful in removing excess helicity from the small scale field that is known to be responsible for the the quenching of dynamos.

4.1 The Large-scale magnetic helicity

The mean field approach to turbulent dynamos is a two scale approach . Therefore, corresponding quantities to those previously defined can also be defined for the mean and the random small-scale fields.

Consider Magnetic helicity defined in (2.8). One can define a corresponding quantity for the mean field in a similar manner to that of the previous case, except that all the quantities will now have an overbar denoting their averages:

$$\overline{H} = \int_V \overline{\mathbf{A}} \cdot (\nabla \times \overline{\mathbf{A}})\, dV = \int_V \overline{\mathbf{A}} \cdot \overline{\mathbf{B}}\, dV$$

Important insights into the working of the turbulent Dynamo can be gained by understanding the time evolution of the helicity densities of the Large scale and the small scale fields. Specifically, the approximate conservation of helicity is a strong constraint that holds in the non-linear evolution of Dynamos. Such constrains provide important means of understanding the non linear Dynamo.

Once again, the time evolution of the net (volume integrated) large scale magnetic helicity can be derived in the same manner as that of the previous definition ((2.9)). First, one begins with the time derivative of the large-scale helicity density:

$$\frac{\partial}{\partial t}\overline{\mathbf{A}}\cdot\overline{\mathbf{B}} = -2\overline{\mathbf{E}}\cdot\overline{\mathbf{B}} - \nabla\cdot(\phi\overline{\mathbf{B}} + \mathbf{E}\times\overline{\mathbf{A}})$$

Now, one substitutes for the Ohm's Law for the mean field instead of the usual Ohm's Law (as in (2.2)):

$$\overline{\mathbf{J}} = \sigma(\overline{\mathbf{E}} + \overline{\mathbf{V}}\times\overline{\mathbf{B}} + \overline{\boldsymbol{\mathcal{E}}})$$

This gives:

$$\frac{\partial(\overline{\mathbf{A}}\cdot\overline{\mathbf{B}})}{\partial t} = 2\overline{\boldsymbol{\mathcal{E}}}\cdot\overline{\mathbf{B}} - 2\eta\overline{\mathbf{B}}\cdot\nabla\times\overline{\mathbf{B}} - \nabla\cdot(\overline{\phi\mathbf{B}} + \overline{\mathbf{E}}\times\overline{\mathbf{A}})$$

The total volume integrated rate of change would be:

$$\frac{\partial\overline{H}}{\partial t} = 2\langle\overline{\boldsymbol{\mathcal{E}}}\cdot\overline{\mathbf{B}}\rangle - 2\eta\langle\mathbf{B}\cdot(\nabla\times\overline{\mathbf{B}})\rangle - \oint_{\partial V}(\overline{\phi\mathbf{B}} + \overline{\mathbf{E}}\times\overline{\mathbf{A}})d\mathbf{S}\cdot\hat{\mathbf{n}}$$

Here the volume integrals are represented by angular brackets.

Let us now look at magnetic helicity for the random small-scale field.

4.2 The Small-scale Helicity

The total volume integrated small scale helicity can be defined in a straightforward manner:

$$\overline{h} = \int_V \overline{\mathbf{a}\cdot(\nabla\times\mathbf{a})}\,dV = \int_V \overline{\mathbf{a}\cdot\mathbf{b}}\,dV \tag{4.1}$$

Now, one can also compute the rate of change of the averaged small-scale helicity by subtracting the above equation from the corresponding equation for total helicity (2.9):

$$\frac{\partial\overline{h}}{\partial t} = -2\langle\overline{\mathscr{E}}\cdot\overline{\mathbf{B}}\rangle - 2\eta\langle\overline{\mathbf{b}}\cdot(\nabla\times\overline{\mathbf{b}})\rangle - \oint_{\partial V}(\overline{\tilde{\phi}\mathbf{b}} + \overline{\mathbf{e}\times\mathbf{b}})d\mathbf{S}\cdot\hat{\mathbf{n}} \tag{4.2}$$

4.3 Conservation of net helicity

Now that the time evolution equation for both the helicities are available, the time evolution of the net helicity can be written down:

$$\frac{\partial}{\partial t}(\overline{\mathbf{A}}\cdot\overline{\mathbf{B}} + \overline{\mathbf{a}\cdot\mathbf{b}}) = -2\eta\langle\mathbf{B}\cdot(\nabla\times\overline{\mathbf{B}})\rangle - 2\eta_T\langle\overline{\mathbf{b}}\cdot(\nabla\times\overline{\mathbf{b}})\rangle - \oint_{\partial V}(\overline{\phi\mathbf{B}} + \overline{\mathbf{E}}\times\overline{\mathbf{A}})\cdot\hat{\mathbf{n}}\,d\mathbf{S} - \oint_{\partial V}(\overline{\tilde{\phi}\mathbf{a}} + \overline{\mathbf{e}\times\mathbf{b}})\cdot\hat{\mathbf{n}}\,d\mathbf{S}$$

This expression says that in ideal MHD, the sum of the mean and the small-scale magnetic helicities is constant in time, and the electromotive force term is responsible for transferring helicity from the small-scale to the large-scale fields.

In general, one would like to understand better the helicity transfer between the large-scale and small- scales,and have a concept of *helicity density*, discuss its time evolution to understand how the large scale and small scale helicities evolve with time and their relation to the working of dynamos. Any appeal to the concept of small-scale helicity density will have to be gauge-invariant and well defined. One such definition for the weakly inhomogeneous case was proposed by Subramanian and Brandenburg (2005,[1]). This is what will be discussed in the following section.

4.4 A Gauge invariant definition for helicity density

It is appealing to define the random small-scale helicity density as the integrand of the expression (4.1):

$$h = \mathbf{a} \cdot (\nabla \times \mathbf{a})$$

However, the main issue with this definition is that, it is not explicitly gauge invariant. Therefore, an alternate definition was proposed (Subramanian, Brandenburg, 2005)[1]

4.4.1 Helicity density through the Gauss Linking Formula

The helicity of the small scale field can be described in a well defined manner through the *Gauss Linking Formula*:

$$h_G = \frac{1}{4\pi} \int \int \mathbf{b}(\mathbf{x}) \cdot \left[\mathbf{b}(\mathbf{y}) \times \frac{\mathbf{x}-\mathbf{y}}{|\mathbf{x}-\mathbf{y}|^3} \right] d^3\mathbf{x}\, d^3\mathbf{y} \tag{4.3}$$

One can define an auxillary field for the random small scale magnetic vector potential:

$$\mathbf{a}_C(\mathbf{x}) = \frac{1}{4\pi} \int \mathbf{b}(\mathbf{y}) \times \frac{\mathbf{x}-\mathbf{y}}{|\mathbf{x}-\mathbf{y}|^3}\, d^3y \tag{4.4}$$

which gives $\nabla \times \mathbf{a}_C(\mathbf{x}) = \mathbf{b}$ and $\nabla \cdot \mathbf{a}_C(\mathbf{x}) = 0$. This also casts the first equation (4.3) in the form of the naive definition (4.1). This is evidently the magnetic vector potential in the Coloumb gauge. The preceding claims can also be justified as follows: Since we are solving for the magnetic vector potential given the magnetic field,

$$\nabla \times \mathbf{a} = \mathbf{b}$$

we can as well use the amperes law that gives the solution (4.4) for the magnetic field given the current as they are related by the same equation as above. As far as the gauge is concerned, we have $\nabla \cdot \mathbf{b} = 0$ which in this case is just the Coloumb gauge for the vector potential $\mathbf{a}$.

Given this definition (4.1) of the small scale helicity, one can apply it in any gauge as long as the fields are closed in the volume of integration. The definition of Magnetic helicity as a measure of the topological property determining the extent of links associated with the vector potential is then more basic than the definition in (4.1).

This point of view then also allows one to define a gauge invariant magnetic helicity density by making use of the presence of two scales in the system : the correlation scale l of the random small scale field and an intermediate scale L such that $l << L << R_s$ where the last is the length-scale of the system. For this, one needs to transform the coordinates from $\mathbf{x}, \mathbf{y}$ to the mean and relative coordinates $\mathbf{r} = (\mathbf{x} - \mathbf{y})$, $\mathbf{R} = (\mathbf{x} + \mathbf{y}/2)$. In the case of weakly inhomogeneous turbulence, all the two point correlation functions of the random small scale field will then vary rapidly with $\mathbf{r}$ and slowly with $\mathbf{R}$. The integral (4.3) can now be written as:

$$h_G = \frac{1}{4\pi} \int d^3\mathbf{R} \int d^3\mathbf{r}\ \varepsilon_{ijk}\, M_{ij}(\mathbf{r}, \mathbf{R})\ \frac{r_k}{r^3}$$

Where $M_{ij}(\mathbf{r}, \mathbf{R}) = \overline{b_i(\mathbf{k_1} b_j \mathbf{k_2})}$.

Now, if $M_{ij} \to 0$ as $r \to L$, then one can still capture the dominant contributions to the integral by integrating only over $\mathbf{r}$. The resulting function that depends exclusively on $\mathbf{R}$ can then be defined as *magnetic helicity density*:

$$h(\mathbf{R}) = \frac{1}{4\pi} \int_{L^3} d^3\mathbf{r}\ \varepsilon_{ijk}\, M_{ij}\ (\mathbf{r}, \mathbf{R})\ \frac{r_k}{r^3} \tag{4.5}$$

Later, letting $L \to \infty$ does not have much effect on the integral due to the two scale condition $l << L << R_s$.

As mentioned earlier, we would like to calculate the time evolution of the small-scale magnetic helicity density. Since the various approximations that are necessary can be done more easily in the Fourier space and due to two-scale approach to the problem, the concept of *Double Fourier Transforms* would be necessary, which is the topic of the upcoming section.

4.4.2 Double Fourier Transforms

In this sub-section, the idea of *double Fourier transforms* would be detailed that would prove to be useful in dealing with the two-scale approach introduced above.

Assume that the convention for the Fourier transform is such that the Fourier transform and its inverse are written as:

$$f(\mathbf{x_i}) = \int d^3\mathbf{k_i}\ \hat{f}(k_i)\ e^{i\mathbf{k}_i \cdot \mathbf{x}_i}$$

$$\hat{f}(k_i) = \frac{1}{(2\pi)^3} \int d^3\mathbf{x_i}\ f(\mathbf{x}_i)\ e^{-i\mathbf{k_i} \cdot \mathbf{x_i}}$$

Consider the double correlations of the random small-scale field. Its general form involving the two functions f and g can be depicted as its inverse Fourier transform over both the coordinate vectors :

$$\overline{f(\mathbf{x_1}) g(\mathbf{x_2})} = \int d^3\mathbf{k_1} \int d^3\mathbf{k}_2\ \hat{\phi}(f, g, \mathbf{k_1}, \mathbf{k_2}) e^{i\mathbf{k_1} \cdot \mathbf{x_1}}\ e^{i\mathbf{k_2} \cdot \mathbf{x_2}}$$

Now, one can use mean-relative coordinates in Fourier space corresponding to the above set of coordinates $(\mathbf{r}, RR)$: $\mathbf{K} = (\mathbf{k_1} + \mathbf{k_2})$, $\mathbf{k} = (\mathbf{k_1} - \mathbf{k_2})/2$, which casts the integral into

$$\overline{f(\mathbf{x_1})g(\mathbf{x_2})} = \int d^3\mathbf{k}\ \hat{\phi}(\hat{f},\hat{g},\mathbf{k},\mathbf{R})e^{i\mathbf{K}\cdot\mathbf{R}}\ e^{i\mathbf{k}\cdot\mathbf{r}}$$

Where

$$\phi(\hat{f},\hat{g},\mathbf{k},\mathbf{R}) = \int d^3\mathbf{k}\ \overline{\hat{f}(\mathbf{k}+\frac{1}{2}\mathbf{K})\hat{g}(-\mathbf{k}+\frac{1}{2}\mathbf{K})} \tag{4.6}$$

The following notation will be used to address the respective correlation functions:

$$\begin{aligned} m_{ij} &= \phi(\hat{b}_i,\hat{b}_j,\mathbf{k},\mathbf{R}) \\ v_{ij} &= \phi(\hat{v}_i,\hat{v}_j,\mathbf{k},\mathbf{R}) \\ \hat{\chi}_{jk} &= \phi(\hat{u}_j,\hat{b}_k,\mathbf{k},\mathbf{R}) \end{aligned}$$

For instance, in this notation the mean electromotive force $\overline{\mathscr{E}} = \varepsilon_{ijk}\int \chi_{jk}(\mathbf{k},\mathbf{R})d^3\mathbf{k}$. The helicity density defined through (4.5) can now be written as:

$$h(R) = \int\int \varepsilon_{ijk}\overline{\hat{b}_i(\mathbf{k}+\frac{1}{2}\mathbf{K})\hat{b}_j(-\mathbf{k}+\frac{1}{2}\mathbf{K})} \times \left(\frac{ik_k}{k^2}\right) e^{i\mathbf{K}\cdot\mathbf{R}}d^3\ \mathbf{k}\ d^3\ \mathbf{K}$$

On the other hand, one can compare this with the helicity density expression one would have arrived at if one had assumed the Cloumb-guage definition $h_C(R) = \mathbf{a}_C\cdot\mathbf{b}$. In this case, $\mathbf{k}$ in the above expression would have been replaced by $(\mathbf{k}+\frac{1}{2}\mathbf{K})$ everywhere. However the two expressions will be identical only when turbulence is homogeneous. This shows that in general, the two helicities, defined through the Coloumb-Gauge and the Gauss-linking formula are not equal.

4.4.3 Time evolution of the small-scale helicity density

We will now calculate the time derivative of the above expression. The time derivative of the integrand is a sum of two terms:

$$\overline{\dot{\hat{b}}_i(\mathbf{k}+\frac{1}{2}\mathbf{K})\hat{b}_j(-\mathbf{k}+\frac{1}{2}\mathbf{K})} + \overline{\hat{b}_i(\mathbf{k}+\frac{1}{2}\mathbf{K})\dot{\hat{b}}_j(-\mathbf{k}+\frac{1}{2}\mathbf{K})} \tag{4.7}$$

Here the dot represents time derivative. Now, use the second of the Maxwell's equation

$$\nabla\times\mathbf{e} = -\frac{\partial\mathbf{b}}{\partial t}$$

written in Fourier space $\dot{\hat{e}}_i = \varepsilon_{ijk}ik_j\hat{b}_k$. Here, the expression for the small-scale electric field is obtained by comparing the above equation with the evolution equation for the small scale magnetic field (3.3). Substituting this, the equation for time evolution of $\overline{h}$ can be written as:

$$\mathbf{e} = -\mathbf{u}\times\overline{\mathbf{B}} - \overline{\mathbf{U}}\times\mathbf{b} - \mathbf{u}\times\mathbf{b} + \overline{\mathbf{u}\times\mathbf{b}} + \eta\mathbf{j} \tag{4.8}$$

Also,change $\mathbf{k}$ to $-\mathbf{k}$ in the term that corresponds to the second term for the time derivative above in the expression (4.7). This will simplify the expression to the following:

$$\frac{\partial h(\mathbf{R})}{\partial t} = \int\int \Big\{ -2\overline{\hat{e}_q(\tfrac{1}{2}\mathbf{K}+\mathbf{k})\hat{b}_q(\tfrac{1}{2}\mathbf{K}-\mathbf{k})} \;+\; 2\frac{K_j k_q}{k^2}\,\overline{\hat{e}_q(\tfrac{1}{2}\mathbf{K}+\mathbf{k})\hat{b}_j(\tfrac{1}{2}\mathbf{K}-\mathbf{k})} \;-\; \frac{K_s k_s}{k^2}\,\overline{\hat{e}_q(\tfrac{1}{2}\mathbf{K}+\mathbf{k})\hat{b}_q(\tfrac{1}{2}\mathbf{K}-\mathbf{k})}\Big\} e^{i\mathbf{K}\cdot\mathbf{R}}\, d^3\mathbf{K}\, d^3\mathbf{k}$$

Let the integral over the three terms be denoted by A_1, A_2 and A_3 respectively. The definition for $\mathbf{e}$ from the Maxwell's equation and from the definition of ϕ as in (4.6), it is evident that

$$A_1 = -2\overline{\mathbf{e}\cdot\mathbf{b}} = 2\overline{\mathbf{b}\cdot(u\times\overline{\mathbf{B}})} - 2\eta\overline{\mathbf{j}\cdot\mathbf{b}} = -2\overline{\mathcal{E}}\cdot\overline{\mathbf{B}} - 2\eta_T\overline{\mathbf{j}\cdot\mathbf{b}}$$

In the above terms inside the integral, only A_1 has no components of $\mathbf{K}$ multiplying it. The other two terms A_2 and A_3 have a factor of K_i multiplying it. Since any multiplicative factor of K_i in the Fourier integral implies a derivative with respective to the corresponding spatial component of $\mathbf{R}$, the latter two terms vanish in homogeneous turbulence. Thus, in homogeneous turbulence, the equation for the time evolution of the small scale magnetic helicity is given by:

$$\frac{\partial h}{\partial t} = -2\overline{\mathcal{E}}\cdot\overline{\mathbf{B}} - 2\eta\overline{\mathbf{j}\cdot\mathbf{b}} \tag{4.11}$$

Comparing this with (4.2)), one finds that this is just a local generalization of its volume integrated counterpart.

Let us now consider the inhomogeneous case where the other two terms A_2 and A_3 are also important. The two terms are evidently scalars that depend on the large scale derivatives of $\mathbf{K}$. These thus appear as divergence of Fluxes in the evolution equation. The only term that acts as a source i.e. a volume generation term is the A_1 term. The evaluation of the helicity fluxes due to the remaining terms upto first order in derivatives of R_i can be done involving a series of manipulations as follows:

First, consider the contribution to the fluxes due to the term containing $\overline{\mathbf{B}}$: i.e. $\mathbf{e} = -\mathbf{u}\times\overline{\mathbf{B}}$. In Fourier space, this equation can be written as:

$$\hat{e}_q(\mathbf{k}) = \varepsilon_{qlm}\int \hat{u}_l(\mathbf{k}-\mathbf{k}')\overline{\mathbf{B}}_{\mathbf{m}}(\mathbf{k}')\, d^3\,\mathbf{k}'$$

One substitutes this into the terms A_2 and A_3 (keeping in mind that the dependence is now on $\mathbf{k}_1$, which is $(\frac{1}{2}\mathbf{K}+\mathbf{k})$). For A_2, the integral expression now looks like:

$$A_2 = -\frac{K_j k_q}{k^2}\int d\mathbf{k}'\, d\mathbf{k}\, d\mathbf{K}\, \varepsilon_{qlm}\overline{u_l((\tfrac{1}{2}\mathbf{K}+\mathbf{k})-\mathbf{k}')b_j((\tfrac{1}{2}\mathbf{K}-\mathbf{k}))}\overline{B}_m(\mathbf{k}')e^{i\mathbf{K}\cdot\mathbf{R}}$$

Now, the variable $\mathbf{K}'$ is used instead of $\mathbf{K}$, which are related by $\mathbf{K}' = \mathbf{K}-\mathbf{k}'$. The equation then takes the form:

$$A_2 = -\frac{(K'_j+k'_j)k_q}{k^2}\int d\mathbf{k}'\, d\mathbf{k}\, d\mathbf{K}\, \varepsilon_{qlm}\hat{\chi}_{lm}(\mathbf{k}-1/2\mathbf{k}',\mathbf{K}')\overline{B}_m(\mathbf{k}')e^{i\mathbf{k}'\cdot\mathbf{R}}\, e^{i\mathbf{K}'\cdot\mathbf{R}}$$

Since we are only interested in terms that are first order in the derivatives of $\mathbf{R}$, we should ignore all terms that have a quadratic dependence on $\mathbf{K}'$ or $\mathbf{k}'$. This implies that when the correlation function is expanded in Taylor series in the variable $\mathbf{k}'$, only the zeroth order term that does not bear derivatives with respect to $\mathbf{k}'$ survives.After this approximation, integrating over $\mathbf{k}'$ and $\mathbf{K}'$, one , the expression can be written as a divergence term:

$$A_2 = -\nabla_j \overline{\mathcal{F}}_j^{VC}$$
$$A_3 = -\nabla_j \overline{\mathcal{F}}_j^{A}$$

Where the corresponding fluxes are given by:

$$\overline{\mathcal{F}}_i^{VC} = 2\varepsilon_{qlm}\overline{\mathbf{B}}_l(\mathbf{R}) \int \frac{ik_q}{k^2}\chi_{mi}\, d^3\, k$$
$$\overline{\mathcal{F}}_i^{A} = -\varepsilon_{qlm}\overline{\mathbf{B}}_l(\mathbf{R}) \int \frac{ik_i}{k^2}\chi_{mq}\, d^3\, k$$

It is evident that $\overline{\mathcal{F}}_i^{A}$ only depends on the antisymmetric part of the cross correlation χ_{mq} whereas $\overline{\mathcal{F}}_i^{VC}$ is sensitive to the symmetric part as well.

There are then the fluxes that arise due to the term containing the mean magnetic field in the expression for the small-scale electric field (i.e. (4.8)). Calculations similar to the one carried out above can also be done for the term containing the mean velocity $\overline{\mathbf{V}} \times \mathbf{b}$, which yields an advective flux i.e. flux proportional to the advective velocity.

$$\overline{\mathcal{F}}_i^{bulk} = \varepsilon_{qlm}\overline{\mathbf{U}}_m(\mathbf{R}) \int \left(2\frac{ik_q}{k^2}m_{li} - \frac{ik_i}{k^2}m_{lq}\right) d^3\, k$$

This becomes even more convincing when one calculates the above Fluxes for the case of isotropic turbulence, which yields $\overline{\mathcal{F}} = h\overline{\mathbf{U}}$.

The only left over terms are due to the mean turbulent emf, which has no contribution and the term $-\mathbf{v} \times \mathbf{b}$ which induces a triple correlation in the flux, which can be denoted by $\mathcal{F}^{triple}$.

In order to further relate these quantities to observables, one needs to evaluate the correlation functions in terms of the Mean field quantities which requires a closure scheme. One such closure scheme is the *Minimal Tau Approximation*, which is thought to be valid even when R_m is not small (see [3] for further details).

The time evolution equation for the magnetic helicity density can now be written as:

$$\frac{\partial \overline{h}}{\partial t} + \nabla \cdot \overline{\mathcal{F}} = -2\overline{\mathcal{E}} \cdot \overline{\mathbf{B}} - 2\eta \overline{\mathbf{j} \cdot \mathbf{b}}$$

Where the flux $\overline{\mathcal{F}}_i = \overline{\mathcal{F}}_i^{VC} + \overline{\mathcal{F}}_i^{A} + \overline{\mathcal{F}}_i^{bulk} + \overline{\mathcal{F}}_i^{triple}$

This is a generalization of the local conservation law for the small-scale helicity (4.11) for the weakly inhomogeneous case. An important fact to note here is that an equation such as the above could not have been derived if a gauge-invariant definition of the small-scale helicity density had not been possible.

The features of this equation can now be summarized as under:

- There are no fluxes if the turbulence is homogeneous.
- $\overline{\mathcal{F}}_i^{VC} = 0$ and $\overline{\mathcal{F}}_i^{A} = 0$ if the turbulence is isotropic but inhomogeneous.
- The VC and A fluxes are proportional to the correlations χ_{ij} and the mean field $\overline{\mathbf{B}}$
- Also, it can be shown that in case of equipartition $v_{ij} = m_{ij}$, the VC and A-fluxes vanish (see [1] for further details).

Quenching

The helicity density evolution equation without the fluxes (4.11) says that if the system is at at steady state, $\overline{\mathcal{E}} \cdot \overline{\mathbf{B}} + \eta \overline{\mathbf{j} \cdot \mathbf{b}} = 0$. This implies that

$$\overline{\mathcal{E}} \cdot \overline{\mathbf{B}} = -\eta \overline{\mathbf{j} \cdot \mathbf{b}} \to 0 \; as \; \eta \to 0$$

which leads to the catastrophic quenching of the tubulent emf parallel to $\overline{\mathbf{B}}$. However, this need not be the case in the presence of helicity fluxes as:

$$\overline{\mathcal{E}} \cdot \overline{\mathbf{B}} = -\frac{1}{2} \nabla \cdot \overline{\mathcal{F}} - \eta \overline{\mathbf{j} \cdot \mathbf{b}}$$

This shows that the magnetic helicity fluxes can alleviate the problem of quenching and are crucial for the proper working of the dynamo.

Chapter 5

Numerical Simulation of Dynamos

In this chapter we present an up-to-date account of the numerical pursuit of solutions to the $\alpha\omega$*Dynamo* in the context of disk galaxies that began towards the end of the project.

5.1 Dynamo Action in Disk Galaxies

5.1.1 The Setup and the Equations

For the application of the Dynamo equations for disk galaxies, one begins by writing down the component equation of (3.2) in the cylindrical coordinate system due to the disk geometry. We know that in disk galaxies, the thickness of the disk is orders of magnitude smaller than the radial extent. Therefore to a good approximation, one can neglect the radial derivatives. Furthermore, with the axisymmetry assumption, the azimuthal derivatives also drop out. The resulting equations describe the local evolution of the magnetic field . For the α^2 Dynamo they are given by:

$$\frac{\partial B_\rho}{\partial t} = -\frac{\partial}{\partial z}(\alpha B_\phi) + \eta \frac{\partial^2 B_\rho}{\partial z^2}$$
$$\frac{\partial B_\phi}{\partial t} = \frac{\partial}{\partial z}(\alpha B_\rho) + \eta \frac{\partial^2 B_\rho}{\partial z^2}$$
$$\frac{\partial B_z}{\partial t} = \eta \frac{\partial^2 B_z}{\partial z^2}$$

R_α , defined as the ratio of the strength of the α term to the diffusive term is:

$$R_\alpha = \frac{\alpha L}{\eta}$$

Where L is a characteristic length-scale. It is usually taken to be the thickness of the disk. Since the z component equation is just a diffusion equation, we will only be interested in the ρ and ϕ component equations.

The boundary conditions corresponding to a disk geometry are that the magnetic field vanishes outside the disk and at the upper and lower boundaries i.e. $B_\rho(\pm h) = 0 = (B_\phi(\pm(z))$ .This can be justified from the Maxwell's equations and the geometry of the disk.

The problem is treated as an initial value problem and the numerical technique employed to implement the code was Runge-Kutta 6th order which can be readily found in the documentation of *Pencil Code* http://pencil-code.nordita.org/.

The technique is implemented in C. The code has been successfully tested for the case of Magnetic diffusion ($\alpha = 0$ .We are currently working on the initial value problem for the α^2 and $\alpha\omega$ dynamo.

Chapter 6

Summary

The differential rotation in galaxies and clusters can amplify fields linearly in time due to the flux freezing effects. However this linear growth is insufficient to account for the present day observed strengths of the large-scale magnetic fields. What one needs is a process that can convert the kinetic energy in the fluid to electromagnetic energy much faster i.e. a Dynamo.

In this project, we reviewed the Turbulent Dynamo and derived its equations from the mean field approach. The Turbulent emf calculated in the First Order Smoothing Approximation results in the well known α effect. Some existing simple analytic solutions to the dynamo equation show that the Dynamo equation can have solutions that are exponential in time. Thus, the tubulent dynamo can potentially amplify the seed fields exponentially.

However, the action of the turbulent dynamo on the large-scale fields creates small-scale helicity, which further leads to a catastrophic quenching of the Dyanmo not apparently observed. A gauge-invariant definition of the small-scale helicity density made possible the calculation of the time evolution of the same, which was shown to lead to certain flux terms alongside the usual conserving terms in the equation (see [1]. These fluxes can then be responsible for the removal of excess helicity in the small scale field, thus alleviating the problem of quenching.

Analytical solutions to the Dynamo equation are difficult to obtain. In order to explore numerical solutions to the Dynamo equation, an Runge-Kutta code (order 6) was implemented in C Programming language to simulate the local evolution of magnetic fields in a disk geometry. These will be relevant to Dynamo action in disk galaxies.

The code has been tested for the case of Magnetic Diffusion. underway. Work on numerically evolving the $\alpha\omega$ dynamo in the context of disk galaxies is currently underway.

Appendix A

Appendix

A.1 The MHD equations

A.1.1 The One Fluid Model

One natural way of beginning with Magnetohydrodynamics is to write down an equation for the evolution of the velocity field of a neutral fluid consisting of ions analogous to the Euler equation for the ideal fluid. The modification would be that there is now an extra force due to the Magnetic field (the Lorentz Force. The force due to the electric field will be zero since the fluid is effectively neutral):

$$\frac{\partial V}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} = -\frac{\nabla p}{\rho_m} - \nabla \phi + \frac{1}{\rho_m}(\rho_q \mathbf{E} + \mathbf{J} \times \mathbf{B})$$

Here $\mathbf{V}$ is the Bulk velocity of the fluid p is the total pressure in the fluid(i.e. accounting for the contribution from all ions/charges), ϕ is the total gravitational potential, ρ_q is the net charge density and $\mathbf{J}$ is the net current density due to motion of all the ions in the fluid.

One then adds by hand the Ohm's law and the Maxwell's equations to complete the description (The continuity equation and the equation of state have been assumed):

Ohm's Law:

$$\mathbf{J} = \rho_q \mathbf{V} + \sigma(\mathbf{E} + \mathbf{V_d} \times \mathbf{B})$$

As is the case usually, one sets $\rho_q = 0$ since for most astrophysical cases of interest the systems being dealt with are effectively electrically neutral. This leaves the term due to advection of the net charge density , i.e. the first term zero in the previous equation. The second term is derived from the fact that the positive and negative ions in a fluid are drifting relative to each other, the Lorentz force due to which balances the force suffered in collisions.

The Maxwell's equations for the plasma can be written as:

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{1}{\varepsilon_0}\rho_q \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \mathbf{B} &= 0 \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{J} - \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} \approx \mu_0 \mathbf{J}
\end{aligned}$$

and the equation of state of the plasma.

A.2 The Two fluid model

The two fluid model gives a more fundamental description of plasmas as being composed of two or more types of charges/ions (eg. electrons and protons). In this approach, one writes down the independent fluid equations corresponding to each component along with an interaction term that arises due to collisions between the species of each kind. Then, one attempts to obtain an equation that describes the the two components of the fluid collectively i.e. as one entity. Through this procedure, one obtains the usual MHD equation of the one fluid model, along with a generalized version of the Ohm's Law (2.2).

A.2.1 Deriving the MHD equation

The easiest way to derive the one fluid equation from the two fluid model is to first consider the 1st moment of the Boltzmann equation for the two species, which yields their respective momentum equations:

Let $f_\alpha(\mathbf{x},\mathbf{v})$ be the phase space distribution function of the αth species (which can be electron and the proton fluids, for instance). The Boltzmann equation for the αth species with collisions, when the external forces are due to Gravity and the Electromagnetic fields is:

$$\frac{df_\alpha}{dt} = \frac{\partial f_\alpha}{\partial t} + \frac{d\mathbf{x}}{dt}\cdot\nabla_{\mathbf{x}} f_\alpha + \frac{d\mathbf{v}}{dt}\cdot\nabla_{\mathbf{v}} f_\alpha = \frac{\partial f_\alpha}{\partial t}|_c \tag{A.1}$$

Where the last term on the right hand side is due to collisions.

This along with the Maxwell's equations :

$$\begin{aligned}
\nabla\cdot\mathbf{E} &= \Sigma_\alpha\, n_\alpha q_\alpha \\
\nabla\times\mathbf{E} &= -\frac{\partial B}{\partial t} \\
\nabla\cdot\mathbf{B} &= 0 \\
\nabla\times\mathbf{B} &= \frac{\partial \mathbf{E}}{\partial t} + \mu_0\Sigma_\alpha\, n_\alpha q_\alpha \mathbf{V}_\alpha
\end{aligned}$$

comprise what is called as the *Vlasov-Maxwell system*. The time derivative of the electric field in the last equation is usually very small and is usually neglected. The collision term is approximated as:

$$\frac{\partial f_\alpha}{\partial t}|_c \approx -\Sigma_\beta\ (\mathbf{V}_\alpha - \mathbf{V}_\beta)\nu_{\alpha\beta}$$

where $\nu_{\alpha\beta}$ is the collision frequency between the involved species.

The 0th moment of this equation yields the usual continuity equation for the αth species:

$$\frac{\partial n_\alpha}{\partial t} + \nabla\cdot(n_\alpha \mathbf{V}_\alpha) = 0$$

The first moment leads to the momentum equation:

$$n_\alpha m_\alpha \left(\frac{\partial}{\partial t} + \mathbf{V}_\alpha \cdot \nabla \right) \mathbf{V}_\alpha = -\nabla p_\alpha - \nabla \phi_\alpha + n_\alpha q_\alpha (\mathbf{E} + \mathbf{V} \times \mathbf{B}) - \int n_\alpha m_\alpha \mathbf{v} \frac{\partial f_\alpha}{\partial t}|_c d\mathbf{v}$$

The one fluid MHD equation can be derived by adding and subtracting the fluid equations for the αth species. This gives:

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0$$
$$\rho_m \frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} = -\nabla \cdot \mathbf{P} - \nabla \phi_\alpha + \rho_q \mathbf{E} + \mathbf{J} \times \mathbf{B}$$

Where the one fluid variables are defined as:

$$\rho_m = \Sigma_\alpha \, n_\alpha m_\alpha$$
$$\rho_q = \Sigma_\alpha \, n_\alpha q_\alpha$$

Center of mass velocity is given by:

$$\mathbf{V}(\mathbf{x}, t) = \frac{\Sigma_\alpha \, n_\alpha m_\alpha \mathbf{V}_\alpha}{\Sigma_\alpha n_\alpha m_\alpha}$$

Current density is:

$$\mathbf{J} = \Sigma_\alpha q_\alpha n_\alpha \mathbf{V}_\alpha$$

The pressure of ions and electrons w.r.t the centre of mass and the total pressure is:

$$\mathbf{P}_{ij}^{CM} = n_\alpha m_\alpha \int (v_i - V_i)(v_j - V_j) f_\alpha d\mathbf{v}$$
$$\mathbf{P} = \Sigma_\alpha \mathbf{P}_\alpha^{CM}$$

Bibliography

[1] *Magnetic Helicity Density and its flux in weakly inhomogeneous turbulence*
The Astrophysical Journal Letters, Volume 648, Number 1. 2006
K. Subramanian, A. Brandenburg

[2] *Principles of Plasma Physics*
Nicholas A. Krall, Alvin W. Trivelpiece

[3] *Astrophysical magnetic fields and nonlinear dynamo theory*
Physics Reports, Volume 417, Issues 1–4, October 2005, Pages 1-209
A. Brandenburg, K. Subramanian,

[4] *Analytic solution of an oscillatory migratory α2 stellar dynamo*
A&A,Volume 598, February 2017, Article Number A117
A. Brandenburg

[5] *Astrophysical Magnetic Fields: from Galaxies to the Early Universe*
Anvar Shukurov, Kandaswamy Subramanian